

25.4 Adjustment sets and the back-door criterion

If confounding is association along back-door paths, the remedy is to block them by conditioning on variables on them, that is, by including those variables as regressors. This section says which sets do the job, proves that they do, and compares their precision.

25.4.1 The criterion

Definition 25.4.1 — Back-door criterion

Let X and Y be nodes of a DAG. A set $\mathcal{Z}$ of nodes, containing neither X nor Y , satisfies the **back-door criterion** relative to (X, Y) if

- (i) no node in $\mathcal{Z}$ is a descendant of X , and
- (ii) $\mathcal{Z}$ blocks every back-door path from X to Y (Definition 25.3.3).

Such a set is called a (back-door) **adjustment set**.

Condition (ii) closes the non-causal routes. Condition (i) protects the causal ones: a descendant of X may be a mediator, whose conditioning removes part of the effect, or a collider, whose conditioning opens a path (Section 25.5). The empty set qualifies exactly when the effect is not confounded, and the parents of X always qualify, provided Y is not one of them, since every back-door path begins $X \leftarrow P$ with P a parent and a non-collider. The graph itself is an assumption that the data cannot check (Exercise (B3)); the criterion turns it into a precise statement about which regression to run.

Recall from Section 25.2 that $Y(x)$ is the value of Y in the model modified by $\text{do}(X = x)$, and that $Y = Y(x)$ on the event $X = x$. Conditional independence of $Y(x)$ and X given $\mathbf{Z}$ is the potential-outcome form of “no confounding given $\mathbf{Z}$ ”, which Rosenbaum and Rubin (1983), adding positivity, called *strong ignorability*.

Theorem 25.4.1 — Back-door adjustment

Let $\mathbf{V}$ follow a structural causal model with graph G , let X and Y be two of its variables, and let $\mathbf{Z}$ be the vector of the variables in a set $\mathcal{Z}$ of other nodes.

- (a) **Adjustment formula.** Suppose X takes finitely many values, $\Pr(X = x \mid \mathbf{Z}) > 0$ with probability one, $E|Y(x)| < \infty$, and $Y(x)$ is conditionally independent of X given $\mathbf{Z}$. Then

$$E(Y \mid \text{do}(X = x)) = E[E(Y \mid X = x, \mathbf{Z})], \quad (25.4.1)$$

where the outer expectation is over the distribution of $\mathbf{Z}$.

- (b) **The criterion gives ignorability.** If $\mathcal{Z}$ satisfies the back-door criterion relative to (X, Y) , then $Y(x)$ is conditionally independent of X given $\mathbf{Z}$, for every x .
- (c) **Linear models.** In a linear structural causal model whose disturbances have

finite positive variances, if $\mathcal{Z}$ satisfies the back-door criterion, the coefficient of X in $L(Y \mid X, \mathbf{Z})$ is the total effect τ of X on Y . Consequently, for independent copies of $\mathbf{V}$, the least squares coefficient of X in the regression of Y on 1, X and $\mathbf{Z}$ converges to τ with probability one.

The proofs of (b) and (c) below are complete when $\mathcal{Z}$ contains every parent of X ; for general adjustment sets they are sketches, since the proof of (c) uses (b).

Proof.

- (a) Using the tower rule, then the conditional independence together with positivity, then $Y = Y(x)$ on $\{X = x\}$,

$$\begin{aligned} E Y(x) &= E[E(Y(x) \mid \mathbf{Z})] \\ &= E[E(Y(x) \mid X = x, \mathbf{Z})] = E[E(Y \mid X = x, \mathbf{Z})], \end{aligned}$$

and $E Y(x) = E(Y \mid \text{do}(X = x))$ by definition.

- (b) *When $\mathcal{Z}$ contains the parents of X .* Let $X = V_k$, and write $\mathbf{U}_{-k}$ for the disturbances other than U_k . A node that is not a descendant of X has no directed path from X , so solving the assignments shows that it is a function of $\mathbf{U}_{-k}$ alone; by (i), so is $\mathbf{Z}$. In the model modified by $\text{do}(X = x)$, the disturbance U_k appears nowhere, so $Y(x)$ is also a function of $\mathbf{U}_{-k}$. Hence U_k is independent of $(Y(x), \mathbf{Z})$, and given $\mathbf{Z}$ it remains independent of $Y(x)$, with its unconditional distribution. Finally $X = f_k(\mathbf{Z}_{\text{pa}}, U_k)$, where $\mathbf{Z}_{\text{pa}}$ is the part of $\mathbf{Z}$ formed by the parents of X . Given $\mathbf{Z}$, X is a function of U_k alone, and so is conditionally independent of $Y(x)$. *General adjustment sets (sketch).* Form a graph containing both the original variables and their counterparts in the model modified by $\text{do}(X = x)$, joined through shared disturbance nodes (a *twin network*); $Y(x)$ is a node of it, and X has no arrows into the counterpart variables. A path from X to $Y(x)$ that leaves X through a child of X must turn back somewhere, at the latest at a disturbance node shared with a counterpart variable, so it meets a collider that is a descendant of X ; by (i) neither that collider nor its descendants are in $\mathcal{Z}$, and the path is blocked. A path that leaves X through a parent starts like a back-door path, and one checks, using (ii) and again (i), that $\mathcal{Z}$ blocks it. So $\mathcal{Z}$ d-separates X from $Y(x)$, and Theorem 25.3.4(a) gives the conditional independence. Pearl (1995, 2009) gives the criterion and its proof in the language of interventions, and Shpitser, VanderWeele and Robins (2010) characterize exactly which sets satisfy Eq. 25.4.1 for every model on G .
- (c) By Proposition 25.3.5(a), $Y = \tau X + R$, where R is the value of Y under $\text{do}(X = 0)$, that is, $R = Y(0)$. The variables R , X and $\mathbf{Z}$ are linear functions of the disturbances with coefficients determined by $\mathbf{B}$, so their covariance matrix depends only on $\mathbf{B}$ and $\mathbf{\Omega}$. The covariance matrix of $(X, \mathbf{Z})$ is positive definite, because $\text{Cov}(\mathbf{V}) = \mathbf{A}\mathbf{\Omega}\mathbf{A}^\top$ with $\mathbf{A}$ invertible and $\mathbf{\Omega}$ positive definite. By Proposition 14.5.2 the coefficient of X in $L(Y \mid X, \mathbf{Z})$ is

$$\frac{\sigma_{YX \cdot \mathbf{Z}}}{\sigma_{XX \cdot \mathbf{Z}}} = \tau + \frac{\sigma_{RX \cdot \mathbf{Z}}}{\sigma_{XX \cdot \mathbf{Z}}},$$

with $\sigma_{XX \cdot \mathbf{Z}} > 0$. Consider the linear model with the same $\mathbf{B}$ and $\mathbf{\Omega}$ but normal disturbances. It has the same covariance matrix of $(R, X, \mathbf{Z})$, hence the same partial covariance $\sigma_{RX \cdot \mathbf{Z}}$. In it $(R, X, \mathbf{Z})$ is jointly normal and, by (b), $R = Y(0)$ is conditionally independent of X given $\mathbf{Z}$, so $\sigma_{RX \cdot \mathbf{Z}} = 0$ by Proposition 3.6.1(b).

Therefore the coefficient is τ . (This step is complete when $\mathbf{Z}$ contains the parents of X , and otherwise inherits the sketch of (b).) The last statement is Proposition 6.11.3. ■

Part (a) averages the conditional mean at $X = x$ over the population distribution of $\mathbf{Z}$, not over its distribution among units with $X = x$ (direct standardization), and needs positivity: where $X = x$ never occurs, the data say nothing about $E(Y | X = x, \mathbf{Z})$. Part (c) answers the question left open after the Frisch–Waugh–Lovell theorem: the multiple-regression coefficient of X in a linear model is the effect of changing X whenever the other regressors form an adjustment set.

Warning. Only the coefficient of X acquires a causal meaning. The coefficients of the adjustment variables are generally not their effects, because X may mediate them; reporting them as effects is the “Table 2 fallacy” (Westreich and Greenland 2013), computed in [Exercise \(B2\)](#).

25.4.2 An example with eight variables

Example. (Choosing an adjustment set)

[Figure 25.4.1](#) shows a linear model with unit-variance disturbances and

$$\begin{aligned} W_2 &= 0.8W_1 + U, & X &= 0.7W_1 + 0.6W_3 + 0.9I + U, & M &= 0.8X + U, \\ Y &= 0.5X + 0.6M + 0.7W_2 + 0.8W_3 + P + U, \end{aligned}$$

with a separate disturbance U in each equation and W_1, W_3, I, P parentless. Summing over the two directed paths, $\tau = 0.5 + 0.8 \times 0.6 = 0.98$. The back-door paths $X \leftarrow W_1 \rightarrow W_2 \rightarrow Y$ and $X \leftarrow W_3 \rightarrow Y$ are blocked by W_1 or W_2 , and by W_3 , and M is a descendant of X . So a set is valid iff it contains W_3 and one of W_1, W_2 , and not M ; I and P are optional.

The listing computes the coefficient of X in $L(Y | X, \mathbf{Z})$ exactly from $\text{Cov}(\mathbf{V}) = \mathbf{A}\mathbf{A}^\top$, with the factor $\sigma_{Y \cdot XZ} / \sigma_{X \cdot Z}$ that governs precision (see below), and the standard deviation over 4000 simulated samples of size $n = 500$:

adjustment set	valid	coefficient of X	$\sigma_{Y \cdot XZ} / \sigma_{X \cdot Z}$	simulated sd
none	no	1.308	1.15	0.052
W_3	no	1.150	1.16	0.052
W_1, W_3	yes	0.980	1.25	0.056
W_2, W_3	yes	0.980	1.06	0.047
W_2, W_3, P	yes	0.980	0.80	0.036
W_1, W_3, I	yes	0.980	1.69	0.076
W_1, W_3, M	no	0.500	1.72	0.077
W_1, W_2, W_3, I, P	yes	0.980	1.17	0.052

Every valid set gives τ ; the invalid sets leave a back-door path open or, with M , keep only the direct effect 0.5. Among valid sets the standard deviation varies by a factor of two (Figure 25.4.2).

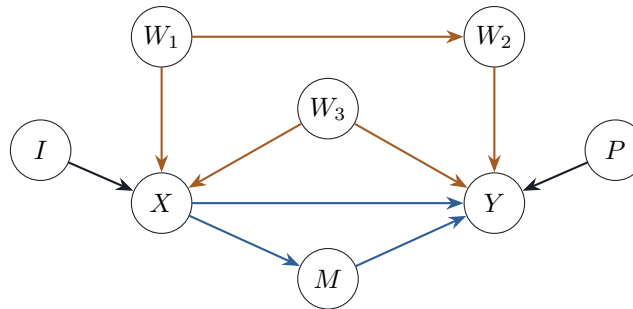


Figure 25.4.1. The graph of [Example \(Choosing an adjustment set\)](#). Blue arrows form the directed paths from X to Y ; brown arrows form the two back-door paths. I influences only X , and P only Y .

```
import numpy as np

names = ["W1", "W3", "I", "P", "W2", "X", "M", "Y"]      # a topological order
k = {v: j for j, v in enumerate(names)}
B = np.zeros((8, 8))                                  # B[j, i]: coefficient of
    i in j's equation
for child, parent, coef in [(("W2", "W1", 0.8), ("X", "W1", 0.7), ("X", "W3", 0.6), (
    "X", "I", 0.9),
                            ("M", "X", 0.8), ("Y", "X", 0.5), ("Y", "M", 0.6), ("Y",
                            "W2", 0.7),
                            ("Y", "W3", 0.8), ("Y", "P", 1.0))]:
    B[k[child], k[parent]] = coef
A = np.linalg.inv(np.eye(8) - B)                      # V = A U
Sigma = A @ A.T                                       # unit disturbance
    variances
tau = A[k["Y"], k["X"]]                             # total effect: sum over
    directed paths

def population_fit(Z):
    """Coefficient of X in L(Y | X, Z) and the large-sample sd factor sigma_{Y.XZ}/
    sigma_{X.Z}."""
    R = [k["X"]] + [k[z] for z in Z]
    coef = np.linalg.solve(Sigma[np.ix_(R, R)], Sigma[R, k["Y"]])
    res_y = Sigma[k["Y"], k["Y"]] - Sigma[k["Y"], R] @ coef
    Zi = [k[z] for z in Z]
    res_x = Sigma[k["X"], k["X"]] - (Sigma[k["X"], Zi] @ np.linalg.solve(Sigma[np.
    ix_(Zi, Zi)], Sigma[Zi, k["X"]]))
    if Z else 0.0)
    return coef[0], np.sqrt(res_y / res_x)

sets = [[], ["W3"], ["W1", "W3"], ["W2", "W3"], ["W2", "W3", "P"], ["W1", "W3", "I"],
    ["W1", "W3", "M"], ["W1", "W2", "W3", "I", "P"]]
print(f"total effect of X on Y: {tau:.3f}")
for Z in sets:
    c, f = population_fit(Z)
    print(f"Z = {{{', '.join(Z)}}}.ljust(26) + f"coefficient {c:6.3f}    sd factor {
    f:5.3f}")
```

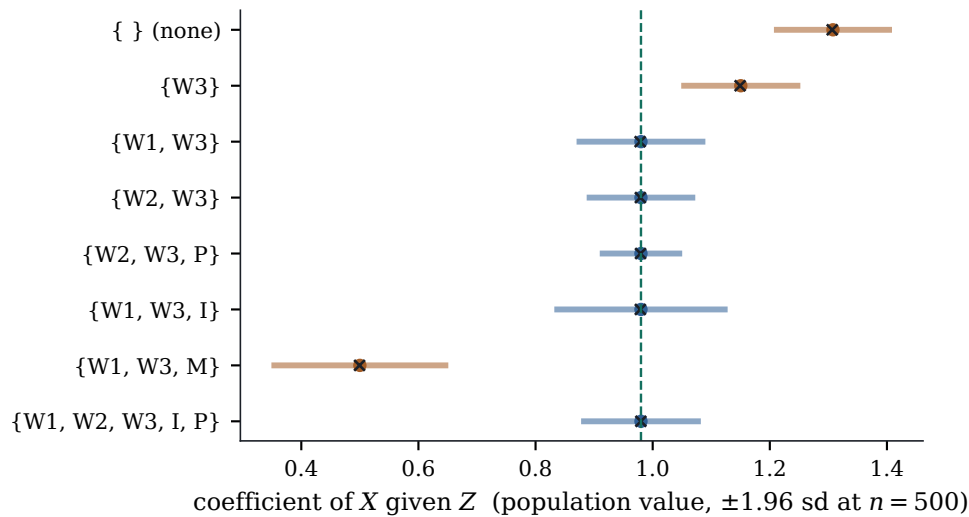


Figure 25.4.1: The coefficient of X given each candidate set in [Example \(Choosing an adjustment set\)](#): population value (dot), a band of ± 1.96 large-sample standard deviations at $n = 500$, and the mean over the simulated samples (cross). Blue sets satisfy the back-door criterion; the dashed line is $\tau = 0.98$.

[Run this code online](#)

25.4.3 Which adjustment set is most precise

All valid sets estimate τ , so the choice among them is a question of variance. With normal disturbances, Y given $(X, \mathbf{Z})$ follows a linear model with variance $\sigma_{Y.XZ}^2$, and given the regressors the coefficient of X has variance $\sigma_{Y.XZ}^2 / \|(\mathbf{I} - \mathbf{M}_Z)\mathbf{x}\|^2$, where $\mathbf{M}_Z$ projects onto the span of $\mathbf{1}$ and the columns of $\mathbf{Z}$ (Theorem 6.6.2 and Eq. 6.6.2). Divided by n , the squared length in the denominator is the residual mean square of the regression of X on $\mathbf{Z}$, which converges to $\sigma_{X.Z}^2$ with probability one (Proposition 6.11.3). So n times the variance tends to

$$v(\mathcal{Z}) = \frac{\sigma_{Y.XZ}^2}{\sigma_{X.Z}^2}, \quad (25.4.2)$$

the square of the factor in the table. A good adjustment set explains much of Y and little of X .

Proposition 25.4.2 — Adding a variable to an adjustment set

Let $\mathcal{Z}$ and $\mathcal{Z} \cup \{W\}$ both satisfy the back-door criterion, with all covariance matrices positive definite.

- (a) If $\sigma_{YW.XZ} = 0$, then $v(\mathcal{Z} \cup \{W\}) \geq v(\mathcal{Z})$.
- (b) If $\sigma_{XW.Z} = 0$, then $v(\mathcal{Z} \cup \{W\}) \leq v(\mathcal{Z})$.

Proof. Partial variances are the mean squared errors of best linear predictors, and projecting onto a larger space can only reduce them (Theorem 2.5.2(c)). (a) By Propo-

sition 14.5.2 applied with $(X, \mathbf{Z})$ as conditioning variables, $\sigma_{YW \cdot XZ} = 0$ means that W has coefficient zero in $L(Y \mid X, \mathbf{Z}, W)$, so this predictor equals $L(Y \mid X, \mathbf{Z})$ and the numerator of Eq. 25.4.2 is unchanged; the denominator can only decrease. (b) Likewise $\sigma_{XW \cdot Z} = 0$ means $L(X \mid \mathbf{Z}, W) = L(X \mid \mathbf{Z})$, so the denominator is unchanged, while the numerator can only decrease.

■

In [Example \(Choosing an adjustment set\)](#), a direct computation from $\text{Cov}(\mathbf{V}) = \mathbf{A}\mathbf{A}^\top$ gives $\sigma_{YI \cdot XW_1W_3} = 0$ and $\sigma_{XP \cdot W_2W_3} = 0$: the pure cause I of X satisfies (a) and the pure cause P of Y satisfies (b). Accordingly, adding I to $\{W_1, W_3\}$ raises the factor from 1.25 to 1.69; adding P to $\{W_2, W_3\}$ lowers it from 1.06 to 0.80. Replacing W_1 by W_2 , closer to Y on the back-door path, also helps. Henckel, Perković and Maathuis (2022) turn these observations into a graphical rule for the valid set of smallest asymptotic variance in linear models: roughly, the parents of Y and of the mediators, other than the mediators and X .

If every adjustment set contains an unrecorded variable, as in [Example \(Clinic visits and health\)](#), other structure must be used: an instrumental variable (Theorem 24.4.2 in [Chapter 24](#)), or the front-door criterion of Pearl (1995), which uses an observed mediator; or a sensitivity analysis asks how strong an unobserved confounder would have to be, using formulas such as Eq. 25.3.3.

25.4.4 Exercises

25.4.4.1 A. Check your understanding

Exercise A1.

In [Example \(Choosing an adjustment set\)](#), decide which of $\{W_1, W_2\}$, $\{W_3, P\}$ and $\{W_2, W_3, M\}$ satisfy the back-door criterion, and explain the coefficients of X that the listing gives for them: 1.201, 1.150 and 0.500.

Solution. None does. $\{W_1, W_2\}$ leaves $X \leftarrow W_3 \rightarrow Y$ open, and $\{W_3, P\}$ leaves $X \leftarrow W_1 \rightarrow W_2 \rightarrow Y$ open, so both coefficients exceed $\tau = 0.98$. The set $\{W_2, W_3, M\}$ blocks both back-door paths but contains the mediator M , violating (i); holding M fixed leaves only the direct arrow, with coefficient 0.5.

25.4.4.2 B. Practice

Exercise B1.

Let $X \in \{0, 1\}$ and let Z take values $1, \dots, K$. In a sample, stratum z has n_z units, a fraction $\hat{p}_z \in (0, 1)$ of them with $X = 1$, and difference of mean responses $\hat{\delta}_z = \bar{y}_{1z} - \bar{y}_{0z}$. Show that the sample version of Eq. 25.4.1 for the effect $E(Y \mid \text{do}(X = 1)) - E(Y \mid \text{do}(X = 0))$ is $\sum_z (n_z/n) \hat{\delta}_z$, while the coefficient of X in the least squares regression of Y on X and the K stratum indicators is

$$\hat{\beta}_X = \frac{\sum_z n_z \hat{p}_z (1 - \hat{p}_z) \hat{\delta}_z}{\sum_z n_z \hat{p}_z (1 - \hat{p}_z)}.$$

When do they agree?

Solution. Replacing $E(Y \mid X = x, Z = z)$ by $\bar{y}_{xz}$ and the distribution of Z by the sample fractions n_z/n gives the first expression. For the second, apply Theorem 6.6.2 with $\mathbf{X}_2$ the stratum indicators. Projecting onto their span replaces each value by its stratum mean (Example (The one-way layout)), so $\tilde{x}_i = x_i - \hat{p}_{z(i)}$. Then $\hat{\beta}_X = \sum_i \tilde{x}_i y_i / \sum_i \tilde{x}_i^2$. Within stratum z , $\sum_i \tilde{x}_i^2 = n_z \hat{p}_z (1 - \hat{p}_z)$ and $\sum_i \tilde{x}_i y_i = n_z \hat{p}_z (1 - \hat{p}_z) (\bar{y}_{1z} - \bar{y}_{0z})$, since the treated contribute $(1 - \hat{p}_z) n_z \hat{p}_z \bar{y}_{1z}$ and the untreated $-\hat{p}_z n_z (1 - \hat{p}_z) \bar{y}_{0z}$. The two agree when the $\hat{\delta}_z$ are all equal or when $\hat{p}_z$ is the same in every stratum. Otherwise the regression weights strata by the variance of X within them, not by their size, and estimates a weighted average effect that is not the average effect (Angrist 1998).

Exercise B2.

In [Example \(Choosing an adjustment set\)](#), show that the coefficient of W_3 in $L(Y \mid X, W_1, W_3)$ is 0.80, the coefficient of the arrow $W_3 \rightarrow Y$, while the total effect of W_3 on Y is 1.388. Which set of variables would one adjust for to estimate the effect of W_3 ?

Solution. Write $Y = 0.5X + 0.6(0.8X + U_M) + 0.7(0.8W_1 + U_{W_2}) + 0.8W_3 + P + U_Y = 0.98X + 0.56W_1 + 0.8W_3 + E$, where $E = 0.6U_M + 0.7U_{W_2} + P + U_Y$ is uncorrelated with X , W_1 and W_3 (none of these depends on the disturbances in E). So $L(Y \mid X, W_1, W_3) = 0.98X + 0.56W_1 + 0.8W_3$. The total effect of W_3 is the sum over its directed paths to Y : 0.8 directly plus 0.6×0.98 through X , which is 1.388. The coefficient in the regression misses the part that goes through X , because X is held fixed. W_3 has no parents and no back-door paths, so the empty set is an adjustment set for its effect: the simple regression of Y on W_3 estimates 1.388.

Exercise B3.

Suppose X is randomized, so it has no parents, and P is a pre-treatment cause of Y . Show that both $\emptyset$ and $\{P\}$ are adjustment sets and that $v(\{P\}) = (1 - \rho_{Y P \cdot X}^2) v(\emptyset)$, where $\rho_{Y P \cdot X}$ is the partial correlation of Y and P given X . Compare with Proposition 18.1.2.